

Trastuzumab in Human Breast Cancer

Overview: Trastuzumab (Herceptin®) is a humanized IgG1 monoclonal antibody approved for the treatment of **HER2-overexpressing (HER2+) breast cancer** in both early (adjuvant/neoadjuvant) and metastatic settings ¹ ². It binds the extracellular domain of the HER2 (ERBB2) receptor on tumor cells, causing receptor downmodulation and blockade of downstream signaling ³. The net effect is inhibition of proliferative pathways (MAPK/ERK, PI3K/AKT) and induction of cell cycle arrest, along with recruitment of immune effector mechanisms (antibody-dependent cellular cytotoxicity, ADCC) to kill HER2+ cells ³ ⁴. In breast cancer, trastuzumab significantly improves disease-free and overall survival in HER2-amplified tumors (e.g. in NSABP B-31/N9831 and HERA trials) and is included in all major guidelines (NCCN, ESMO) for HER2+ disease ¹.

Identifiers & Synonyms

- **ChEMBL:** ChEMBL1201585 ⁵
- **DrugBank:** DB00072 ⁵
- **Synonyms:** GB221; RHUMAB HER2; trastuzumab-anns; trastuzumab-dkst; trastuzumab-dttb; trastuzumab-herw; trastuzumab-pkrb; trastuzumab-qyyp; trastuzumab-strf; trastuzumab-zerc; etc. ⁶
- **Brand/Trade Names:** Herceptin (intravenous), Herceptin Hylecta (subcutaneous hyaluronidase formulation), plus multiple biosimilars (e.g. Herzuma, Kanjinti, Ogivri, Ontruzant, Trazimera, Zercepac) and combination products (e.g. Phesgo, a pertuzumab+trastuzumab SC formulation) ⁷ ⁸.

Mechanism of Action

Trastuzumab's activity is mediated by binding to the extracellular domain of the **HER2/ERBB2** receptor on breast cancer cells ². This binding blocks HER2 receptor cleavage and induces **receptor downmodulation**, thereby interrupting HER2-driven signaling ³. As a result, downstream proliferative cascades (notably the **MAPK/ERK** and **PI3K/AKT** pathways) are inhibited ³. For example, trastuzumab treatment leads to upregulation of the cyclin-dependent kinase inhibitor p27^Kip1, enforcing G1 cell cycle arrest ⁹. Overall, trastuzumab reduces tumor cell proliferation and can induce apoptosis. In addition to direct signaling blockade, trastuzumab's Fc region recruits immune effector cells: it binds Fcγ receptors on natural killer cells and macrophages, mediating **antibody-dependent cell-mediated cytotoxicity (ADCC)** against HER2+ tumor cells ¹⁰ ⁴. Trastuzumab has also been shown to inhibit tumor angiogenesis ⁴. Together, these mechanisms underlie trastuzumab's potent anti-tumor effects in HER2+ breast cancer.

Primary Targets (Human)

- **ERBB2 (HER2)** (HGNC:3430) – the direct binding target of trastuzumab ².

- (Via Fc region) **FCGR2A** (HGNC:3616) and **FCGR3A** (HGNC:3619) – low-affinity Fcγ receptors on immune cells, engaged to mediate ADCC (note: these are not tumor targets, but mediate the immune mechanism of trastuzumab).

Pathways (Overview)

Trastuzumab impacts a network of pathways crucial to HER2+ breast cancer cell survival, proliferation, and immune interactions. Key pathways (with representative database identifiers) include:

- **ErbB/HER2 Signaling:** (Reactome: *R-HSA-1227986*; KEGG: *hsa04012*; MSigDB: *KEGG_ERBB_SIGNALING_PATHWAY*, *HALLMARK_HER2_SIGNALING*). This pathway encompasses HER2 heterodimerization and activation of downstream effectors. Trastuzumab blocks this pathway at the receptor level ³.
- **PI3K/AKT/mTOR Signaling:** (Reactome: *R-HSA-2219528*; KEGG: *hsa04151*; MSigDB: *HALLMARK_PI3K_AKT_MTOR_SIGNALING*). A key survival pathway downstream of HER2. Inhibition by trastuzumab reduces tumor survival signals ³. Conversely, activating mutations in PIK3CA or loss of PTEN reactivate PI3K/AKT and confer trastuzumab resistance ¹¹.
- **MAPK/ERK Signaling:** (Reactome: *R-HSA-5683057*; KEGG: *hsa04010*; MSigDB: *HALLMARK_KRAS_SIGNALING_UP/DN*). Another HER2-driven growth pathway. Trastuzumab inhibits this cascade ¹². Upregulation of MAPK signaling (e.g. via RAS/RAF mutations) is associated with reduced response.
- **Cell Cycle Regulation:** (Reactome: *R-HSA-1640170*; KEGG: *hsa04110*; MSigDB: *HALLMARK_G2M_CHECKPOINT*, *REACTOME_CELL_CYCLE*). Trastuzumab induces G1 arrest by increasing CDK inhibitors (e.g. p27^Kip1) ⁹ and decreasing cyclin/CDK activity. This slows proliferation. Loss of cell-cycle checkpoints (e.g. downregulation of p27) underlies resistance ¹³.
- **Apoptosis:** (Reactome: *R-HSA-109581*; KEGG: *hsa04210*; MSigDB: *HALLMARK_APOPTOSIS*). Pro-apoptotic pathways can be engaged by HER2 blockade. Trastuzumab can sensitize cells to apoptotic death. In resistance, anti-apoptotic signaling (e.g. via STAT3) is upregulated ¹⁴.
- **Fcγ Receptor Signaling / ADCC:** (KEGG: *hsa04666* FcγR-mediated phagocytosis; GO: *GO:0038094* “Fc-gamma receptor signaling”). Trastuzumab-bound cancer cells trigger FcγR signaling in NK cells/macrophages, leading to ADCC ¹⁰. Effective ADCC is a key part of its MoA.
- **Natural Killer (NK) Cell Mediated Cytotoxicity:** (KEGG: *hsa04650*; MSigDB Immune gene sets). Related to ADCC, involves perforin/granzyme release. Strong NK activation correlates with trastuzumab sensitivity.
- **Angiogenesis (VEGF):** (Reactome: e.g. *R-HSA-381426* VEGF signaling; KEGG: *hsa04370*). Trastuzumab has been shown to reduce tumor angiogenesis ⁴.
- **Notch / STAT3 Signaling:** (Reactome: *R-HSA-1640470* Signaling by NOTCH; *R-HSA-9031628* STAT3 activation). Not a direct target of trastuzumab, but tumor cells often upregulate Notch–STAT3

pathways in response to HER2 blockade. Trastuzumab treatment can increase Jagged1/Notch1 activity and STAT3-mediated survival signals ¹⁵, driving resistance via cancer stem cell and inflammatory pathways.

- **JAK/STAT (IL-6) Pathway:** (KEGG: *hsa04630* JAK-STAT signaling; Reactome: related). IL-6/JAK-STAT signaling is implicated in resistance (often downstream of Notch/STAT3) and immune regulation.
- **Growth Factor Cross-Talk:** Other receptor tyrosine kinases (EGFR/HER1, IGF1R, MET) can engage bypass pathways. Although not directly inhibited by trastuzumab, increased signaling through these can undermine trastuzumab effects (resistance pathways).

Detailed Pathway Regulation

- **Upregulated by Trastuzumab:** In HER2+ tumor cells, trastuzumab upregulates **cell-cycle arrest and pro-apoptotic pathways**, including p27^{Kip1}-mediated G1 checkpoint activation ⁹. Immune effector pathways (NK-cell activation, FcγR signaling) are effectively “up” in the presence of trastuzumab-bound targets (reflecting enhanced ADCC) ¹⁰. Note that in some contexts, *compensatory* pathways become upregulated after HER2 blockade: e.g. Jagged1/Notch and STAT3 signaling often increase following trastuzumab, promoting survival (a resistance mechanism ¹⁵).
- **Downregulated by Trastuzumab:** The primary action is *downregulation* of HER2-driven signaling: MAPK/ERK and PI3K/AKT cascades are suppressed ³, as is downstream cell-cycle progression (cyclin D/E and CDK activity fall). Angiogenic signaling is also reduced ⁴. Broadly, expression of proliferation-associated genes (E2F targets, etc.) is decreased.
- **Sensitivity Pathways (Upregulated):** Tumors with **robust immune signatures** respond better. For instance, high FCGR3A (CD16) expression on NK cells or high tumor-infiltrating lymphocytes correlates with trastuzumab sensitivity (effective ADCC) ¹⁰. High baseline activity of apoptotic pathways and CDK inhibitors (e.g. p27) also predicts response ⁹.
- **Sensitivity Pathways (Downregulated):** Trastuzumab is most effective when intrinsic proliferative signaling is low at baseline. For example, tumors without activating PIK3CA mutations or with intact PTEN (i.e. *low* PI3K/AKT pathway activity) are more sensitive, because they are less able to bypass HER2 blockade ¹¹. Similarly, low expression of cyclin/E (G1/S drivers) favors sensitivity (since cells are more easily arrested by trastuzumab).
- **Resistance Pathways (Upregulated):** Key resistance mechanisms involve *upregulation* of bypass/survival pathways. Constitutive activation of **PI3K/AKT/mTOR** (via PIK3CA mutations or PTEN loss) drives growth despite HER2 inhibition ¹¹. Activation of **HER3/ERBB3**, EGFR, or IGF1R signaling provides alternate proliferation signals. **SRC kinase** and **STAT3** pathways (often downstream of these RTKs or of Notch/IL-6) are frequently upregulated in resistant cells ¹¹ ¹⁵. Notch signaling (Jagged1/Notch1/STAT3 axis) becomes active after trastuzumab and promotes stemness, proliferation and angiogenesis ¹⁵.
- **Resistance Pathways (Downregulated):** Resistance is also associated with *downregulation* of the very processes trastuzumab induces: resistant cells often have lost p27 expression and have

hyperactive CDK2, allowing cell cycle progression ¹³. Apoptotic signaling is suppressed (e.g. via upregulated BCL2/IL-6) so cells evade death. Furthermore, reduced FcγR expression or function (e.g. certain FCGR polymorphisms) diminishes ADCC, lowering trastuzumab efficacy ¹⁰.

Subtype-Stratified Evidence

- **HER2-Positive (ERBB2-amplified) Breast Cancer:** Trastuzumab is effective **only** in tumors overexpressing HER2 (approx. 15–20% of breast cancers ¹⁶). It improves outcomes in both hormone receptor-positive (often “Luminal B HER2+”) and hormone receptor-negative HER2+ tumors ¹. For early-stage HER2+ disease, trastuzumab is added to standard adjuvant chemotherapy, significantly reducing recurrence. In metastatic HER2+ BC, trastuzumab (with taxane±pertuzumab) is a first-line therapy ¹⁷.
- **HER2-Negative Subtypes (TNBC, Luminal A/B without HER2):** Trastuzumab is **not indicated** because the target is absent. Clinical trials show *no benefit* in pure HER2-negative disease. New approaches (e.g. trastuzumab-deruxtecan in “HER2-low” tumors) are under study ¹⁸, but original trastuzumab requires HER2 amplification (IHC 3+ or FISH+) for use.
- **BRCA1/2-Mutated Cancers:** There is no specific role for trastuzumab based on BRCA status. HER2+ cancers should be treated with HER2-targeted therapy regardless of BRCA, and BRCA-mutant TNBC are managed with platinum/PARP inhibitors rather than trastuzumab.

Contraindications and Safety

Contraindications: None absolute ¹⁹.

Cardiotoxicity: The most serious toxicity is cardiac dysfunction. Trastuzumab can cause asymptomatic LVEF decline, cardiomyopathy, congestive heart failure and cardiac death ²⁰. The risk is highest when given with anthracyclines. Guidelines mandate baseline and periodic cardiac monitoring; trastuzumab should be **withheld** if LVEF falls ≥ 10 –16% below baseline, and **permanently discontinued** if dysfunction persists ²¹.

Infusion Reactions: Trastuzumab infusions can trigger fever, chills, dyspnea and hypotension. Serious anaphylactic reactions (bronchospasm, angioedema) have occurred ²². Infusion rate should be slowed or stopped if reactions occur.

Embryo-Fetal Toxicity: Trastuzumab causes oligohydramnios and fetal harm when given during pregnancy ²³. Women of childbearing potential must use effective contraception during and for 7 months after treatment.

Other: Pulmonary toxicity (interstitial pneumonitis, acute respiratory distress) and serious neutropenia (especially when combined with myelosuppressive chemo) have been reported ²⁴ ²⁵. No known absolute renal or hepatic contraindications exist, but patients with pre-existing lung disease or uncontrolled hypertension should be closely monitored.

Guidelines and Trial Context

Trastuzumab is a cornerstone of guideline-recommended therapy for HER2+ breast cancer. For early-stage disease, NCCN/ESMO recommend 1 year of trastuzumab (with chemotherapy) after surgery. Landmark trials (e.g. NSABP B-31/N9831, HERA) showed ~50% improvement in 5-year DFS with adjuvant trastuzumab ¹. In metastatic HER2+ BC, first-line regimens include trastuzumab+taxane (e.g. paclitaxel) ¹⁷, often with pertuzumab (CLEOPATRA trial). Trastuzumab as monotherapy is used after failure of chemotherapy ¹⁷. Current algorithms also include trastuzumab emtansine (T-DM1) for residual disease (KATHERINE) or progression (EMILIA) in HER2+ cases.

Additional Notes

- **Receptor Variants & Resistance:** A truncated p95-HER2 fragment (lacking the trastuzumab epitope) occurs in ~20–30% of HER2+ tumors and is associated with poor response ²⁶. Alterations such as MUC4 overexpression or HER2 nuclear localization can also prevent antibody binding.
- **Combination Therapies:** Trastuzumab is synergistic with certain chemotherapies (taxanes) and other HER2-directed agents (pertuzumab, lapatinib). Subcutaneous formulations (trastuzumab+hyaluronidase; trastuzumab+pertuzumab) improve convenience ⁸.
- **Biomarkers:** Patients with activating PIK3CA mutations or low NK-cell FCGR expression tend to be less responsive. High levels of tumor-infiltrating lymphocytes and presence of the favorable FCGR3A V158F polymorphism have been linked to better outcomes (ADCC).
- **Drug Administration:** Standard dosing is 8 mg/kg loading IV then 6 mg/kg Q3W, or 4+2 mg/kg weekly. A subcutaneous 600 mg fixed dose exists (with hyaluronidase) ⁸. Typical duration is 1 year in adjuvant use; extension beyond 1 year is not beneficial ²⁷.

Pathway Evidence Table

Pathway (database IDs)	Regulation by Trastuzumab	Sensitivity/ Resistance Association	Biological Rationale	Evidence
ErbB/HER2 signaling (Reactome R-HSA-1227986; KEGG hsa04012)	↓ (inhibited)	Sensitivity ↑ when HER2 is targetable; Resistance if HER2 signaling persists (e.g. p95-HER2)	Trastuzumab binds HER2, inducing receptor downmodulation and blocking downstream signals ³ . High HER2 expression is required for effect.	³ ²

Pathway (database IDs)	Regulation by Trastuzumab	Sensitivity/ Resistance Association	Biological Rationale	Evidence
PI3K/AKT/mTOR (R-HSA-2219528; KEGG hsa04151)	↓	Resistance when ↑ (PIK3CA mutation, PTEN loss); Sensitivity when ↓	Normally activated downstream of HER2; its inhibition leads to decreased growth. Constitutive PI3K/AKT activation drives trastuzumab resistance ¹¹ .	³ ¹¹
MAPK/ERK (R-HSA-5683057; KEGG hsa04010)	↓	Resistance when ↑ (e.g. RAS mutation); Sensitivity when ↓	HER2 blockade reduces Ras/Raf/ERK signaling. Cells require MAPK signaling for proliferation; trastuzumab downregulates this cascade ¹² .	¹² ²⁸
Cell Cycle (G1/S) (R-HSA-1640170; KEGG hsa04110)	↑ (arrest)	Resistance when cell-cycle inhibitors lost; Sensitivity when arrest ↑	Trastuzumab induces G1 arrest via p27 ^{Kip1} upregulation ⁹ , halting proliferation. Loss of p27/↑ CDK2 leads to resistance ¹³ .	⁹ ¹³
Apoptosis (intrinsic/extrinsic) (R-HSA-109581; KEGG hsa04210)	↑ (promoted)	Resistance when anti-apoptotic signals ↑ (e.g. STAT3/BCL2); Sensitivity when apoptosis ↑	HER2 inhibition can activate apoptosis (e.g. via p53, death receptors). Trastuzumab can sensitize cells to apoptotic death. Resistance pathways (Notch/STAT3) upregulate anti-apoptotic signals ¹⁴ .	¹⁴ (anti-apoptotic via Notch)
FcyR signaling / ADCC (KEGG hsa04666; GO: 0038094)	Engaged (immune)	Sensitivity when active (strong ADCC); Resistance if impaired (FCGR polymorphisms)	Trastuzumab-bound cells trigger Fcy receptors on NK cells, leading to cytotoxic killing ¹⁰ . Effective immune engagement enhances tumor clearance.	¹⁰

Pathway (database IDs)	Regulation by Trastuzumab	Sensitivity/ Resistance Association	Biological Rationale	Evidence
NK cell cytotoxicity (KEGG hsa04650)	Engaged (immune)	Similar to ADCC above	Involves perforin/granzyme-mediated killing of trastuzumab-labeled cells. Robust NK response correlates with response.	(Related to ADCC; no direct ref)
Angiogenesis (VEGF) (Reactome e.g. R-HSA-381426; KEGG hsa04370)	↓	Resistance when angiogenesis ↑; Sensitivity when ↓	Trastuzumab inhibits tumor angiogenesis ⁴ , starving tumor. High VEGF-driven angiogenesis may counteract this effect.	⁴
Notch/STAT3 signaling (Reactome R-HSA-1640470; R-HSA-9031628)	↑ in resistance	Resistance when ↑; Sensitivity when ↓	Not directly targeted by trastuzumab. After HER2 blockade, Jagged1/Notch1 and STAT3 become upregulated, driving cell proliferation, stemness and survival ¹⁵ . Inhibiting Notch/STAT3 can restore sensitivity.	¹⁵
JAK/STAT (IL-6) pathway (KEGG hsa04630)	↑ in resistance	Resistance when IL-6/JAK-STAT ↑	Often activated downstream of Notch/STAT3 or inflammatory cytokines, promoting survival. (No direct citation)	(Implied by Notch/STAT3 data)
Cell adhesion/EMT (GO/MSigDB terms)	–	Resistance when EMT ↑	Not a direct trastuzumab effect, but HER2 blockade can influence EMT markers. EMT may be linked to stem-like, trastuzumab-resistant phenotype.	(No direct citation)

References: Key points above are supported by studies of trastuzumab's effects and resistance in human cells ³ ⁹ ¹¹ ¹⁵ ⁴, as well as authoritative drug labels ²⁰ ²³ and pathway databases (KEGG, Reactome, MSigDB).

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